

PORT SCANNER PROJECT SUMMARY

This project is a **Python-based TCP Port Scanner** developed to identify open ports on a target system. The scanner uses Python's socket library to attempt TCP connections on a specified range of ports and reports which ports are open.

Initially, the project started as a basic port scanning script that accepted a target IP/domain and a range of ports, then scanned each port sequentially. During development, the original code had a small syntax error due to an extra single quotation mark (') at the end of the final print statement, which caused the program to fail with a syntax error. After correcting this issue, the scanner worked properly.

The initial version also had several limitations such as plain terminal output, no exception handling, minimal user interface, and lack of professional presentation.

To improve the tool, several enhancements were added:

- **Fixed syntax error in the original code**
- **Professional ASCII PORT SCANNER banner**
- **Colored terminal output using Colorama**
- **Automatic terminal screen clearing**
- **Exception handling for invalid hostnames, network errors, and manual interruption**
- **Cleaner scan status messages**
- **Improved overall user experience**

The target used for testing was a **Metasploitable vulnerable machine** with IP address: **192.168.142.130**

This project demonstrates practical cybersecurity concepts such as:

- Network reconnaissance
- Port enumeration
- TCP socket programming
- Service exposure discovery
- Basic security tool development

Previously Error in the Code:

```
(kali@kali)-[~/Desktop/cybersecurity-automation/Day-01-Port-Scanner]
$ python scanner.py
Enter Target IP/Domain: https://testaspnet.vulnweb.com/
Enter Start Port: 1
Enter End Port: 100

Scanning Target: https://testaspnet.vulnweb.com/

Traceback (most recent call last):
  File "/home/kali/Desktop/cybersecurity-automation/Day-01-Port-Scanner/scanner.py", line 14, in <module>
    result = s.connect_ex((target, port))
socket.gaierror: [Errno -2] Name or service not known
```

```
14     result = s.connect_ex((target, port))
```

New Fixed & Enhanced Code:

```
1 import socket
2 import os
3 from colorama import Fore, init
4
5 init(autoreset=True)
6
7 # Clear terminal
8 os.system('cls' if os.name == 'nt' else 'clear')
9
10 # PORT SCANNER Banner
11 banner = f"""
12 {Fore.RED}
13 PORT
14
15
16
17
18
19
20 SCANNER
21
22
23
24
25 {Fore.YELLOW}
26 """
27
28 """
29
30 print(banner)
31
32 target = input(Fore.WHITE + "Enter Target IP/Domain: ")
33 start_port = int(input("Enter Start Port: "))
34 end_port = int(input("Enter End Port: "))
35
36 print(Fore.CYAN + f"\nScanning Target: {target}")
37 print(Fore.YELLOW + "-" * 50)
```

```

38
39 for port in range(start_port, end_port + 1):
40     try:
41         s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
42         s.settimeout(0.5)
43
44         result = s.connect_ex((target, port))
45
46         if result == 0:
47             print(Fore.GREEN + f"[+] Port {port} is OPEN")
48
49         s.close()
50
51     except KeyboardInterrupt:
52         print(Fore.RED + "\nScan stopped by user.")
53         exit()
54
55     except socket.gaierror:
56         print(Fore.RED + "Hostname could not be resolved.")
57         exit()
58
59     except socket.error:
60         print(Fore.RED + "Could not connect to server.")
61         exit()
62
63 print(Fore.CYAN + "\nScan Completed.")

```

OUTPUT:



```

PORT SCANNER
Enter Target IP/Domain: https://testaspnet.vulnweb.com/
Enter Start Port: 1
Enter End Port: 100
Scanning Target: https://testaspnet.vulnweb.com/
Hostname could not be resolved.

```

PORT SCANNER

```
10 for port in range(start_port, end_port + 1):
11     s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
Enter Target IP/Domain: 192.168.142.130
Enter Start Port: 1
Enter End Port: 100
14 result = s.connect_ex((target, port))
15
[+] Port 21 is OPEN
[+] Port 22 is OPEN
[+] Port 23 is OPEN
[+] Port 25 is OPEN
[+] Port 53 is OPEN
[+] Port 80 is OPEN
20
Scan Completed.
```